MA 301
Test 1
Spring - 2013

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
6		10
7		10
8		10
Score		80

Name: ____________________

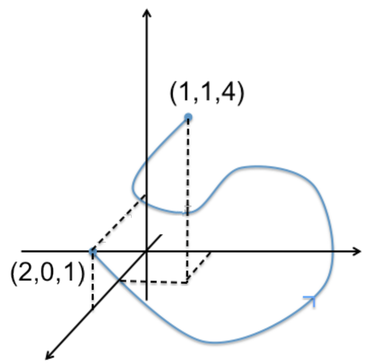
HR: ____________________

Signature: ____________________

READ BEFORE YOU START
* Show all work.
* Change your calculator mode to RADIANS.
* Sloppy solutions may be penalized.
* Use calculators ONLY for computations.
* You must write up every problem in a clear and organized way.
* If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Compute $\oint_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$ if $\mathbf{F} = [x^2 + 4y, x + y^2]$ and C is the rectangle in the $xy-$plane with $0 \leq x \leq 4$ and $-1 \leq y \leq 2$

Problem 2: Let $\mathbf{F} = \left[\frac{y^2}{x}, 2y \ln x, 3z^2\right]$ compute the line integral over C shown below. Justify your work.



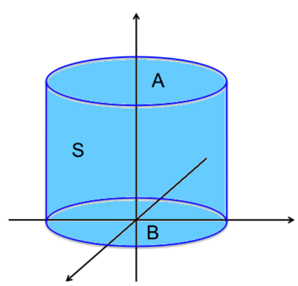
Problem 3: Let $\mathbf{F} = [-y, x]$ and C be a simple closed curve in the $xy-$plane. Let R be the region inside C.

(a) Show that Area$(R) = \frac{1}{2} \oint_C \mathbf{F} \cdot d\mathbf{r}$

(b) Use part (a) to find the area inside the ellipse C: $4x^2 + y^2 = 4$ (*Do NOT use a double integral to get the area*)

Problem 4: **Set-up** the integral that gives the flux of water through the parabolic cylinder $S: y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$ if the velocity flow of water is $\mathbf{v} = [3z^2, 6, 6xz]$ (units: m/s). (water density: $\rho = 1\, kg/m^3$).

Problem 5: The closed circular cylinder shown below has a height of 10 and a base radius of 6. Let A be the top, B the bottom, and S the side surface of the cylinder. Also, let $\mathbf{F} = (7x + 1)\mathbf{i} - 2y\mathbf{j} + z\mathbf{k}$. Find the outward flux through side S. (*Note*: *Vol of a cylinder* $= \pi r^2 h$ and *Area of a circle* $= \pi r^2$)



Problem 6: Let $\mathbf{F} = [x, y, z]$. Compute the flux of $\mathbf{F}$ through the following surfaces S:

(a) S is the hemi-sphere $x^2 + y^2 + z^2 = 25$, $z \geq 0$ with outward orientation. (**Set-up Only**)

(b) S is the surface of the rectangular box: $|x| \leq 1$, $|y| \leq 3$, and $0 \leq z \leq 2$. (**Compute this one**)

Problem 7: **Set-up** the integrals needed to compute the following:

(a) The work done by $\mathbf{F} = [xy, 2y^2]$ in moving a mass along $\mathbf{r}(t) = [\cos t, \sin t]$ & $0 \leq t \leq \frac{\pi}{2}$.

(b) The mass of a wire parameterized by $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$ where $0 \leq t \leq 2$ with density $\delta(x, y, z) = 3\sqrt{5 + x}$.

(c) The surface area of S with parameterization: $\mathbf{r}(u, v) = [u, v, u^3]$, $0 \leq u \leq 1$, $-2 \leq v \leq 2$.

Problem 8: Answer the following true/false questions

T / F	The line integral of any vector field **F** around any closed curve C is zero
T / F	If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$ for one particular closed path, the **F** is path-independent.
T / F	If **F** is path-independent, then there is a potential function for **F**
T / F	If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \neq 0$ for some closed path C, then **F** is path-dependent.
T / F	If $\mathbf{F} = \nabla f$ then **F** is conservative.
T / F	If **F** is path-independent and C is any closed curve, then $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$
T / F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\mathbf{F} = [x, y, z]$ then $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$ is positive
T / F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA = 0$, then $\mathbf{F} = \mathbf{0}$
T / F	If **F** and **G** are vector-fields satisfying div(**F**) = div(**G**), then $\mathbf{F} = \mathbf{G}$
T / F	If **F** is a vector-field such that div(**F**) = 1 and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$ is equal to the volume enclosed by S.